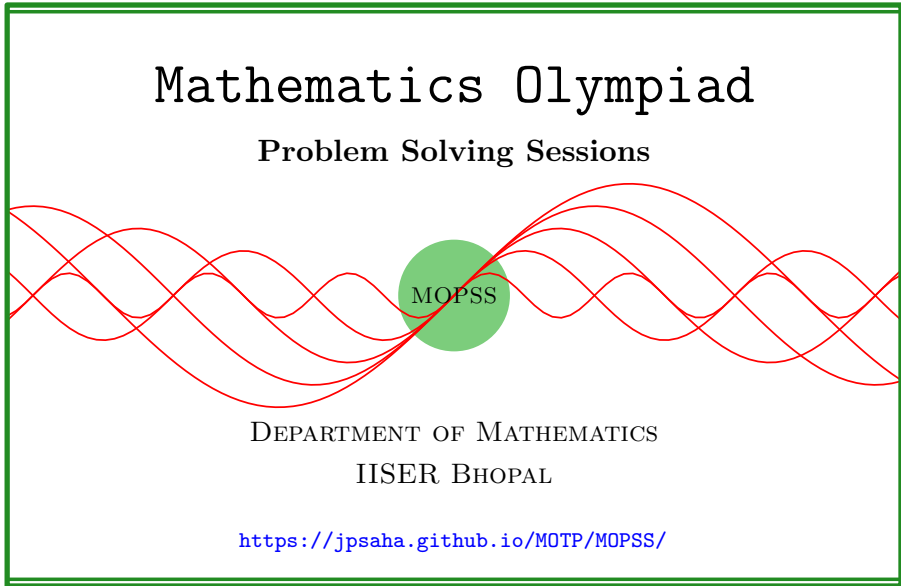


Viète's relations

MOPSS



Suggested readings

- Evan Chen's advice [On reading solutions](https://blog.evanchen.cc/2017/03/06/on-reading-solutions/), available at <https://blog.evanchen.cc/2017/03/06/on-reading-solutions/>.
- Evan Chen's [Advice for writing proofs/Remarks on English](https://web.evanchen.cc/handouts/english/english.pdf), available at <https://web.evanchen.cc/handouts/english/english.pdf>.
- [Notes on proofs](#) by Evan Chen from [OTIS Excerpts](#) [[Che25](#), Chapter 1].
- [Tips for writing up solutions](https://www.math.utoronto.ca/barbeau/writingup.pdf) by Edward Barbeau, available at <https://www.math.utoronto.ca/barbeau/writingup.pdf>.
- Evan Chen discusses why [math olympiads](#) are a valuable experience for [high schoolers](#) in the post on [Lessons from math olympiads](#), available at <https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/>.

List of problems and examples

1.1	Example (India RMO 2012e P2)	2
-----	------------------------------	---

§1 Viète's relations

Example 1.1 (India RMO 2012e P2). cf. [GA17, Problem 141] Let $P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ be a polynomial of degree $n \geq 3$. Knowing that $a_{n-1} = -\binom{n}{1}$ and $a_{n-2} = \binom{n}{2}$, and that all the roots of P are real, find the remaining coefficients. Note that $\binom{n}{r} = \frac{n!}{(n-r)!r!}$.

Solution 1. Let $\alpha_1, \alpha_2, \dots, \alpha_n$ denote the roots of P . Note that

$$\begin{aligned}
 & \sum_{1 \leq i, j \leq n, i \neq j} (\alpha_i - \alpha_j)^2 \\
 &= 2(n-1) \sum_{1 \leq i \leq n} \alpha_i^2 - 2 \sum_{1 \leq i, j \leq n, i \neq j} \alpha_i \alpha_j \\
 &= 2(n-1) \sum_{1 \leq i \leq n} \alpha_i^2 - 4 \sum_{1 \leq i < j \leq n} \alpha_i \alpha_j \\
 &= 2(n-1) \left(\sum_{1 \leq i \leq n} \alpha_i \right)^2 - 4(n-1) \left(\sum_{1 \leq i < j \leq n} \alpha_i \alpha_j \right) - 4 \sum_{1 \leq i < j \leq n} \alpha_i \alpha_j \\
 &= 2n^2(n-1) - 4n \sum_{1 \leq i < j \leq n} \alpha_i \alpha_j \\
 &= 2n^2(n-1) - 4n \binom{n}{2} \\
 &= 0.
 \end{aligned}$$

Since $\alpha_1, \dots, \alpha_n$ are real, it follows that they are all equal. Using $\alpha_1 + \cdots + \alpha_n = n$, we get

$$\alpha_1 = \alpha_2 = \cdots = \alpha_n = 1.$$

This implies that $a_i = (-1)^i \binom{n}{i}$ for any $0 \leq i \leq n-1$. ■

Solution 2. Let $\alpha_1, \alpha_2, \dots, \alpha_n$ denote the roots of $P(x)$. Note that

$$\begin{aligned}
 & (\alpha_1 - 1)^2 + (\alpha_2 - 1)^2 + \cdots + (\alpha_n - 1)^2 \\
 &= \alpha_1^2 + \alpha_2^2 + \cdots + \alpha_n^2 + n - 2(\alpha_1 + \alpha_2 + \cdots + \alpha_n) \\
 &= (\alpha_1 + \alpha_2 + \cdots + \alpha_n)^2 - 2 \sum_{1 \leq i < j \leq n} \alpha_i \alpha_j + n - 2(\alpha_1 + \alpha_2 + \cdots + \alpha_n) \\
 &= n^2 - n(n-1) + n - 2n \\
 &= 0.
 \end{aligned}$$

So the roots $\alpha_1, \alpha_2, \dots, \alpha_n$ are all equal to 1. This implies that $P(x) = (x-1)^n$, and hence a_i is equal to $(-1)^i \binom{n}{i}$ for any $0 \leq i \leq n$. ■

References

- [Che25] EVAN CHEN. *The OTIS Excerpts*. Available at <https://web.evanchen.cc/excerpts.html>. 2025, pp. vi+289 (cited p. 1)
- [GA17] RĂZVAN GELCA and TITU ANDREESCU. *Putnam and beyond*. Second. Springer, Cham, 2017, pp. xviii+850. ISBN: 978-3-319-58986-2; 978-3-319-58988-6. DOI: [10.1007/978-3-319-58988-6](https://doi.org/10.1007/978-3-319-58988-6). URL: <https://doi.org/10.1007/978-3-319-58988-6> (cited p. 2)